

Perturbation de Navier Stokes pour la formation d'ondes capillaires dans une couche mince

restart

$A := 1 :$

$\rho := 2.5e3 : C := 0.74E3 \cdot \rho : \eta := 0.1 \cdot \exp\left(\frac{1E4 \cdot a}{T} + b\right) : Tm := 1970 : kb := 1.38e-23 : g$
 $:= 9.81 : a := 6.2388e0; b := -14.6668$

6.2388

-14.6668 (1)

$\sigma 0 := 0.267 : \sigma 1 := 1.54e-5 :$

$\text{sigma} := T \rightarrow \sigma 0 + \sigma 1 \cdot T$

$T \rightarrow \sigma 0 + \sigma 1 T$ (2)

$\# \kappa 0 := 0.14e2 : \kappa 1 := 0e0 : \# \text{Bauerle}$

$\text{kappa} := T \rightarrow 7.06418e-10 \cdot T^3 - 3.96976e-6 \cdot T^2 + 7.56664e-3 \cdot T + 0.633319e0$

$T \rightarrow 7.06418 \cdot 10^{-10} T^3 + (-1) \cdot 0.00000396976 T^2 + 0.00756664 T + 0.633319$ (3)

(4)

$\text{Intensity} := \frac{\text{fluence}}{\tau} : \text{fluence} := 5e4 : \tau := 300e-15; \# k := 10^{\text{indice}K}; \# \text{gam}R := 10^{\text{indice}Gam};$

$\# k := \frac{2 \cdot \pi}{1e-6 \cdot \Lambda}$

$3.00 \cdot 10^{-13}$ (5)

$T := 'T' : h := 'h' :$

(6)

$k := 'k' : T := 'T' : h := 'h' : \text{period2} := T \rightarrow$

$$- \sqrt{\frac{\eta k^2 \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)}}{\rho^2 ((e^{kh})^2 + 1)}} \sqrt{2}$$

$$- \frac{2 \sinh(2 k h)}{2 \sinh(2 k h)}$$

$$+ \left(\frac{\left| \frac{d}{dT} \sigma(T) \right| \cdot A \cdot \text{Intensity}}{\text{kappa}(T)} k^2 (g h \rho + \sigma(T) k^3) \sqrt{\frac{\kappa(T) k^2 ((e^{kh})^2 + 1) \rho}{C \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)}}} \right)$$

$$\begin{aligned}
& \left(4 \rho \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)} \left(\frac{1}{((e^{kh})^2 + 1) \rho} ((e^{kh})^2 \right. \right. \\
& \left. \left. - 1) (g h \rho + \sigma(T) k^3) \right) - \frac{\left| \frac{d}{dT} \sigma(T) \right| \cdot A \cdot Intensity}{\kappa(T)} k^2 \right) \\
& \left. \rho \left(1 + \sqrt{\frac{\eta C}{\rho \kappa(T)}} \right) \right) \\
T \rightarrow & - \frac{1}{2} \frac{\sqrt{\frac{\eta k^2 \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)}}{\rho^2 ((e^{kh})^2 + 1)}} \sqrt{2}}{\sinh(2 k h)} \\
& + \frac{1}{4} \left(\left| \frac{d}{dT} \sigma(T) \right| A Intensity k^2 (g h \rho \right. \\
& \left. + \sigma(T) k^3) \right. \\
& \left. \sqrt{\frac{\kappa(T) k^2 ((e^{kh})^2 + 1) \rho}{C \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)}} \sqrt{2} ((e^{kh})^2 + 1)} \right) \\
& \left(\kappa(T) \rho \sqrt{((e^{kh})^2 + 1) ((e^{kh})^2 - 1)} \sqrt{\rho (g h \rho + \sigma(T) k^3)} \left(1 / \right. \right. \\
& \left. \left. ((e^{kh})^2 + 1) \rho) ((e^{kh})^2 - 1) (g h \rho + \sigma(T) k^3) \right) \right)
\end{aligned} \tag{7}$$

$$-\frac{\left|\frac{d}{dT}\sigma(T)\right|AIntensityk^2}{\kappa(T)\rho\left(1+\sqrt{\frac{\eta C}{\rho\kappa(T)}}\right)}\right)$$

$$T0 := 2000e0; k := \frac{2e0 \cdot \text{Pi}}{\text{Lambda}}; \#Intensity := \frac{1e4}{100e-15};$$

$$2000.$$

$$\frac{2 \cdot \pi}{\Lambda} \quad (8)$$

$$\text{evalf}(\text{eval}(\text{eval}(\text{eval}(\text{period2}(T), T = T0)), h = 200e-9), \text{Lambda} = 5000e-9))$$

$$-2.375908125 \cdot 10^{11} \quad (9)$$

$$\text{Lambda} := 10^{\text{LogLambda}}; h := 10^{\text{LogH}}$$

$$10^{\text{LogLambda}}$$

$$10^{\text{LogH}} \quad (10)$$

$$\text{resolution} := 100; \text{maxLogH} := -6; \text{maxLogLambda} := -5$$

$$100$$

$$-6$$

$$-5 \quad (11)$$

$$T0 := 300e0$$

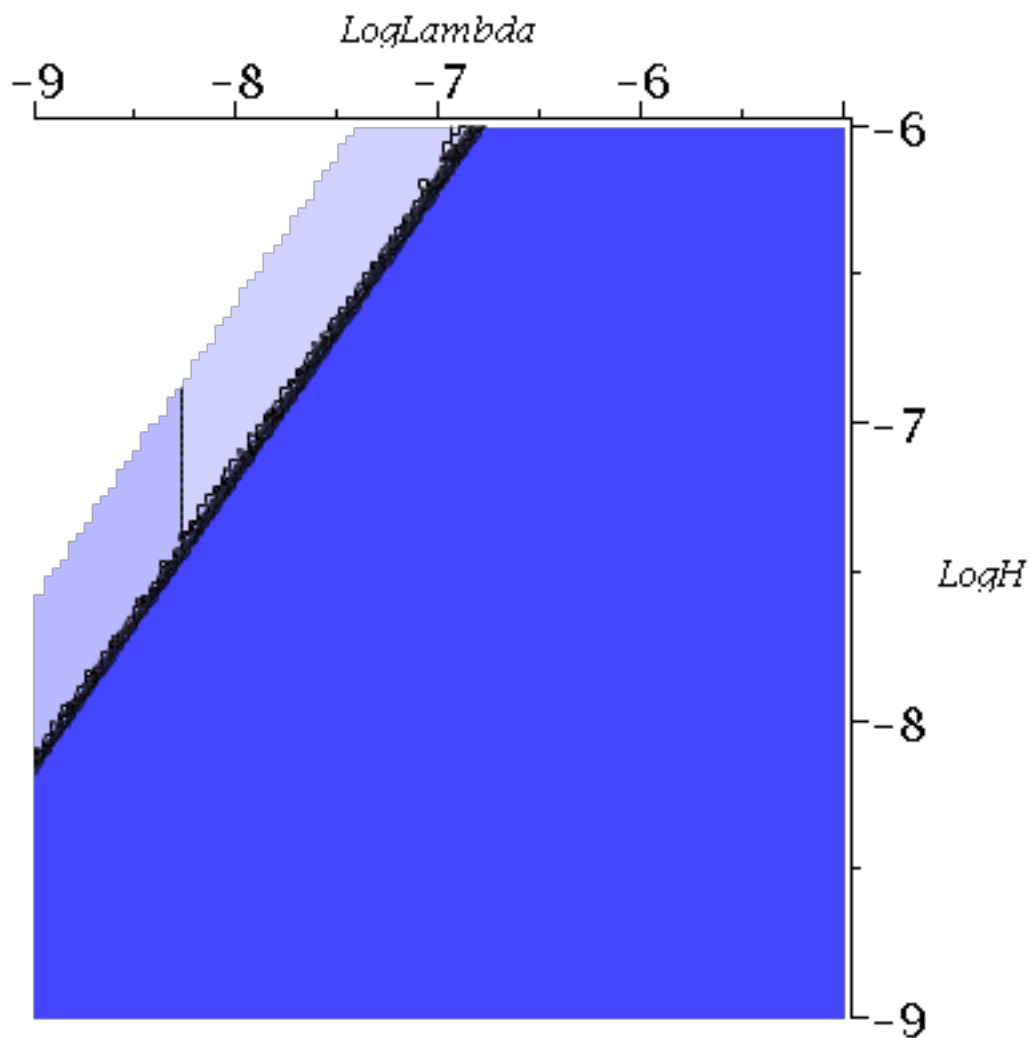
$$300. \quad (12)$$

$$\text{with}(\text{plots}) : \text{contourplot}(\log10(\text{abs}(\text{eval}(\text{period2}(T), T = T0))), \text{LogLambda} = -9$$

$$\text{..maxLogLambda}, \text{LogH} = -9 \text{..maxLogH}, \text{grid} = [\text{resolution}, \text{resolution}], \text{filledregions}$$

$$= \text{true}, \text{contours} = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], \text{coloring} = [\text{white},$$

$$\text{blue}], \text{font} = [\text{TIMES}, 14, \text{REGULAR}])$$

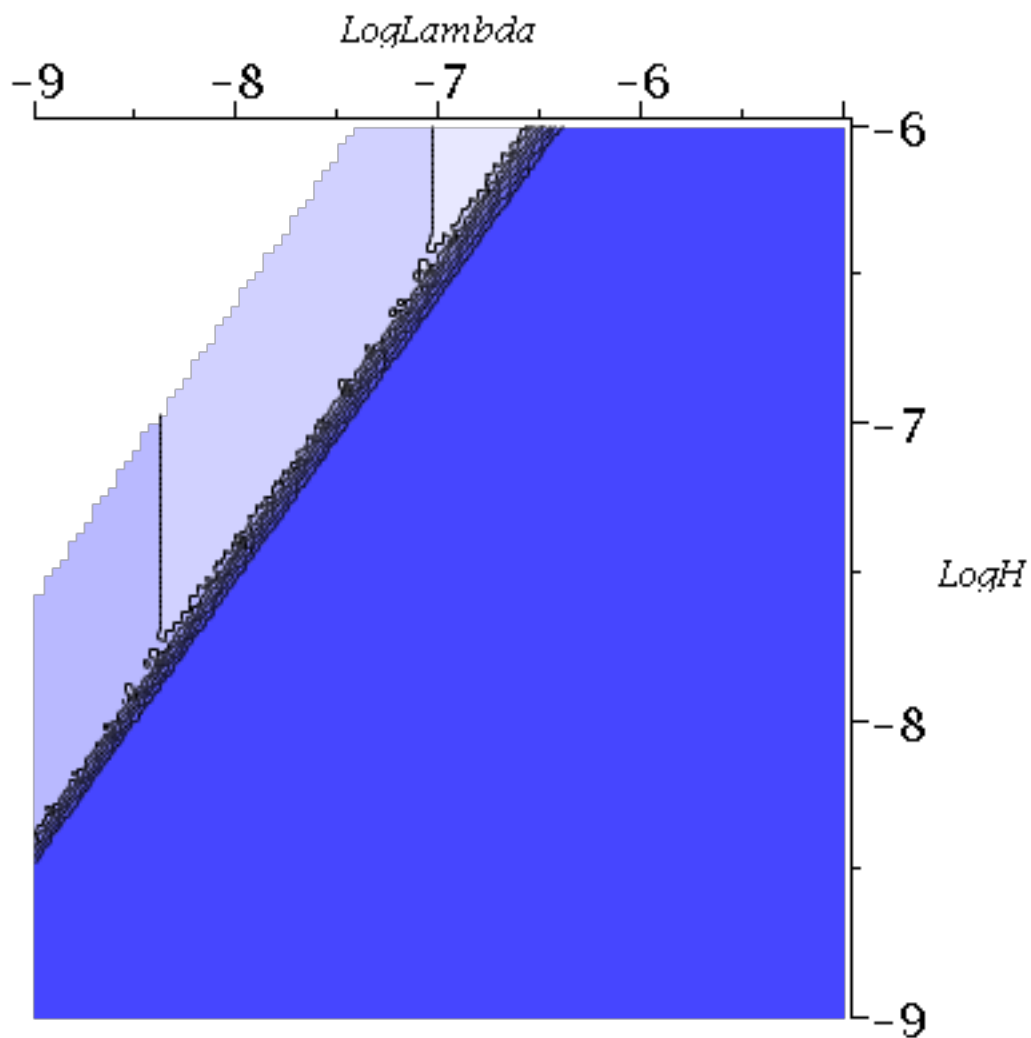


$T0 := 500e0$

500.

(13)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9
    ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```

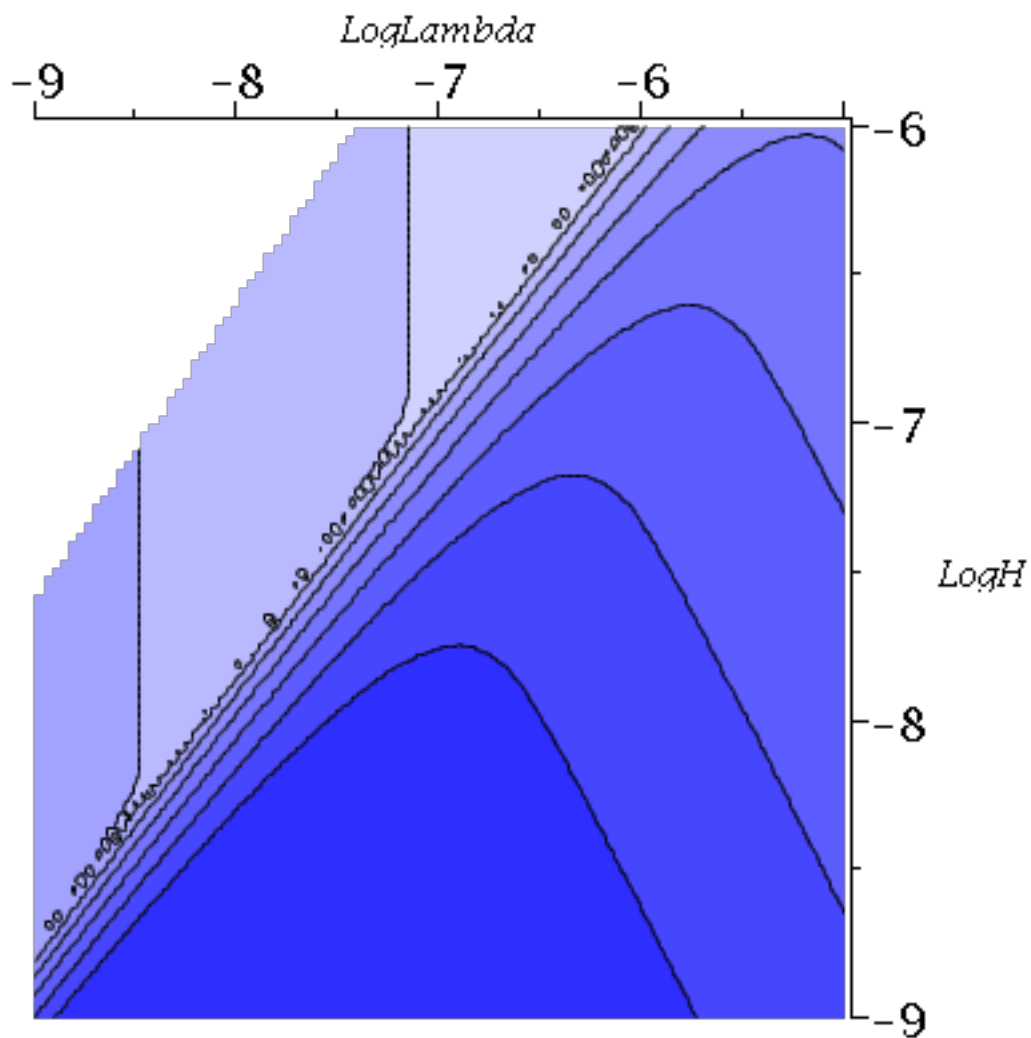


$T0 := 1000e0$

1000.

(14)

```
with(plots): contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9
    ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```

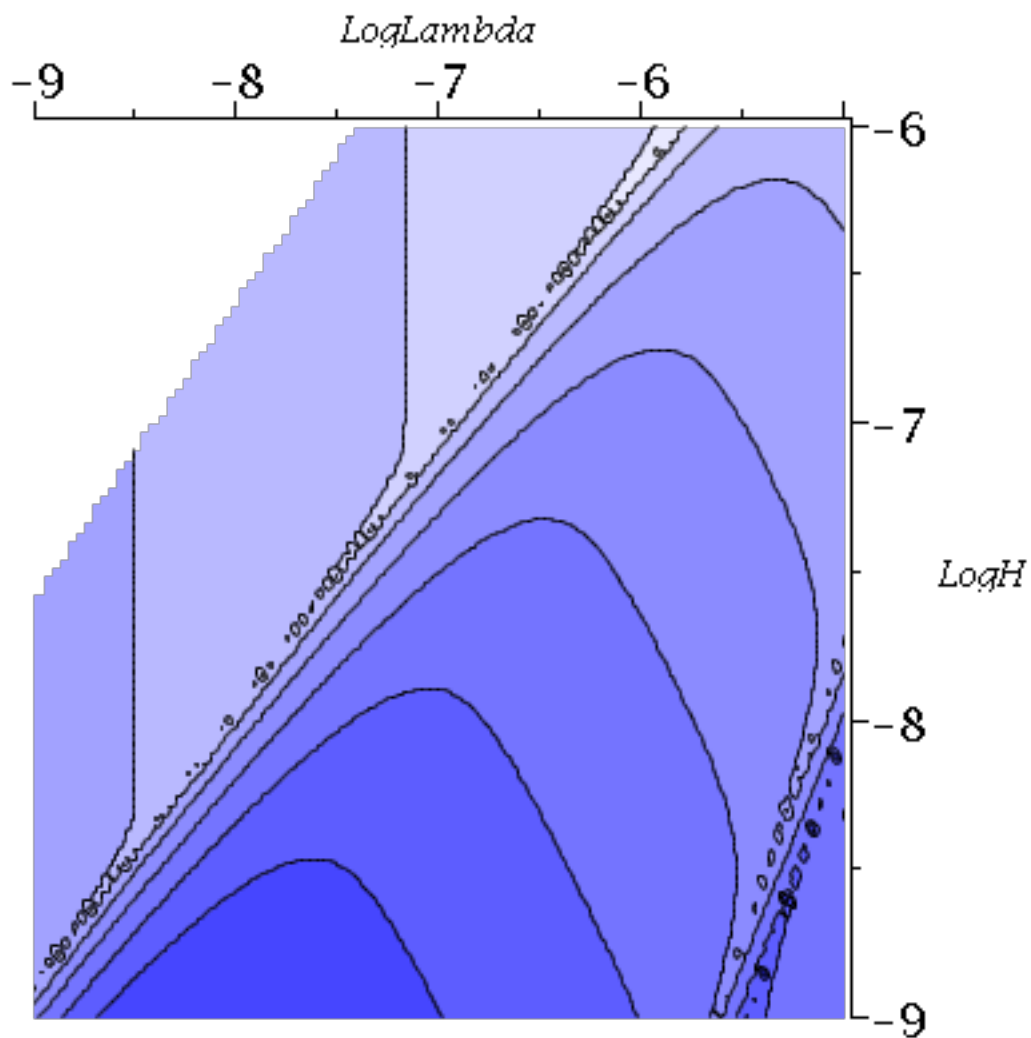


$T0 := 1200e0$

1200.

(15)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9
    ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```

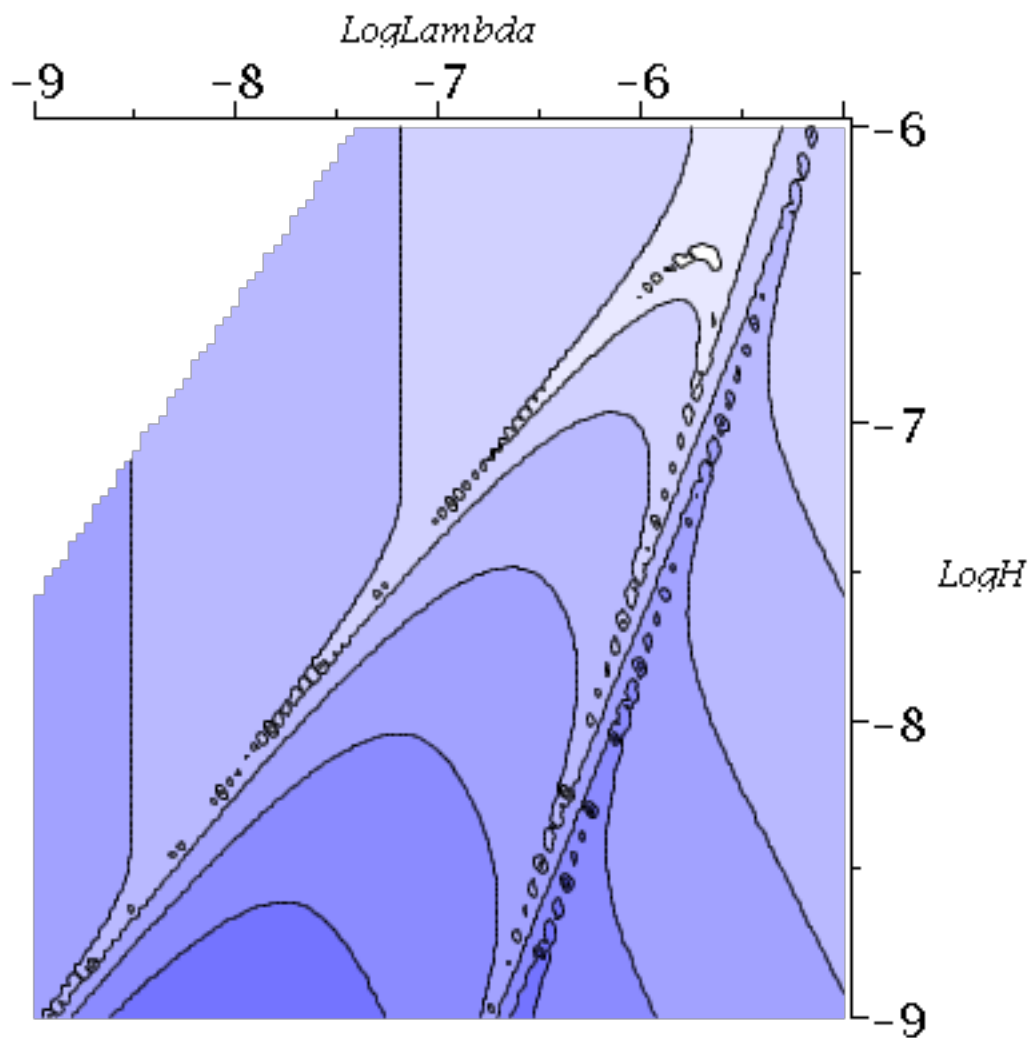


$T0 := 1500e0$

1500.

(16)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9
    ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```

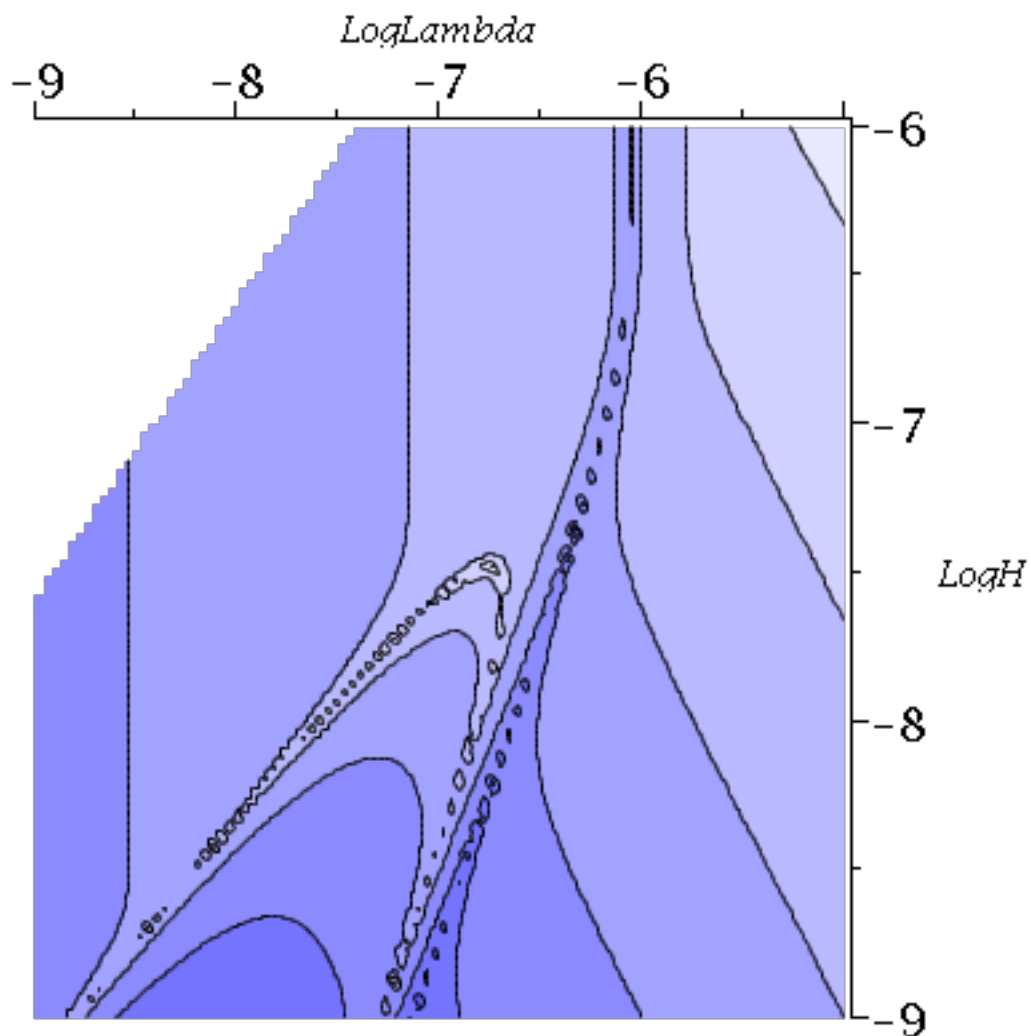


$T0 := 1700e0$

1700.

(17)

with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9 ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white, blue], font = [TIMES, 14, REGULAR])

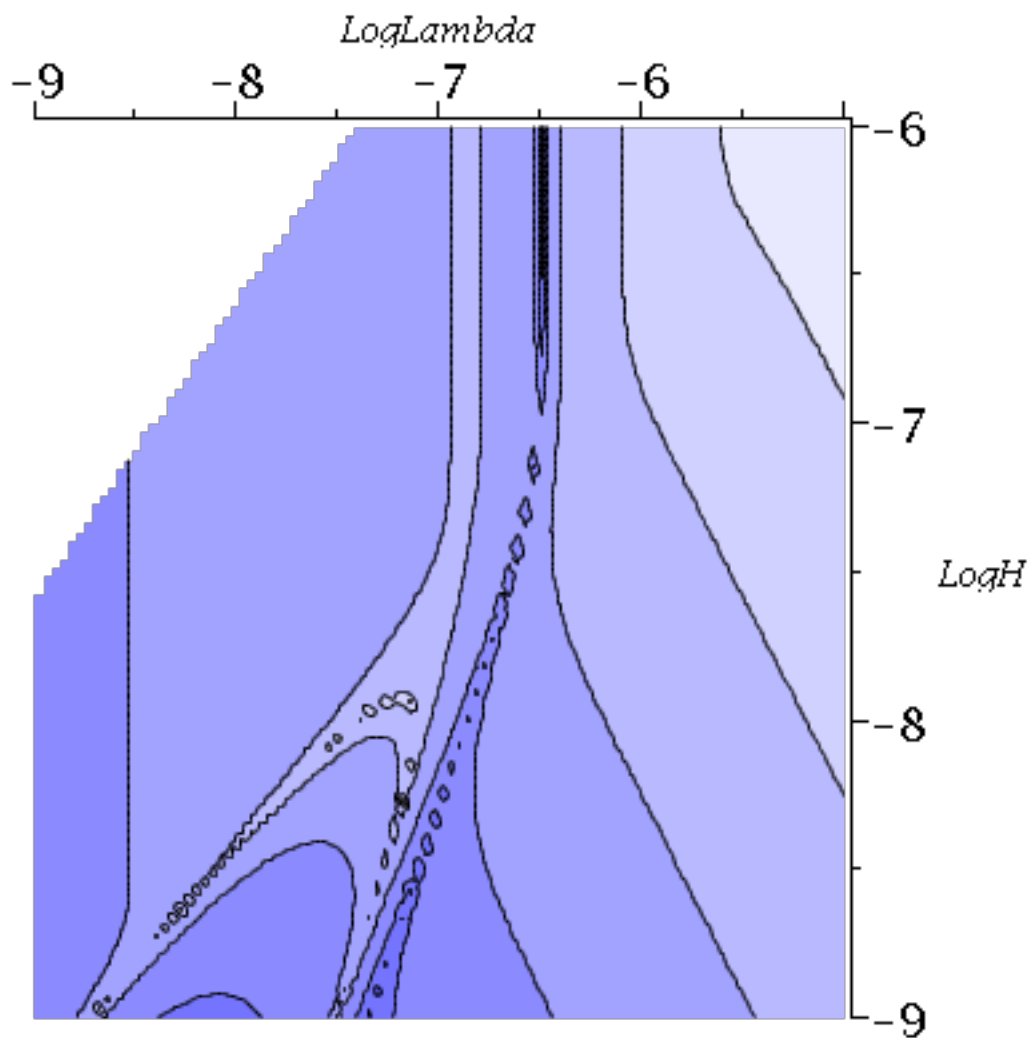


$T0 := 1800e0$

1800.

(18)

with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9 ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white, blue], font = [TIMES, 14, REGULAR])

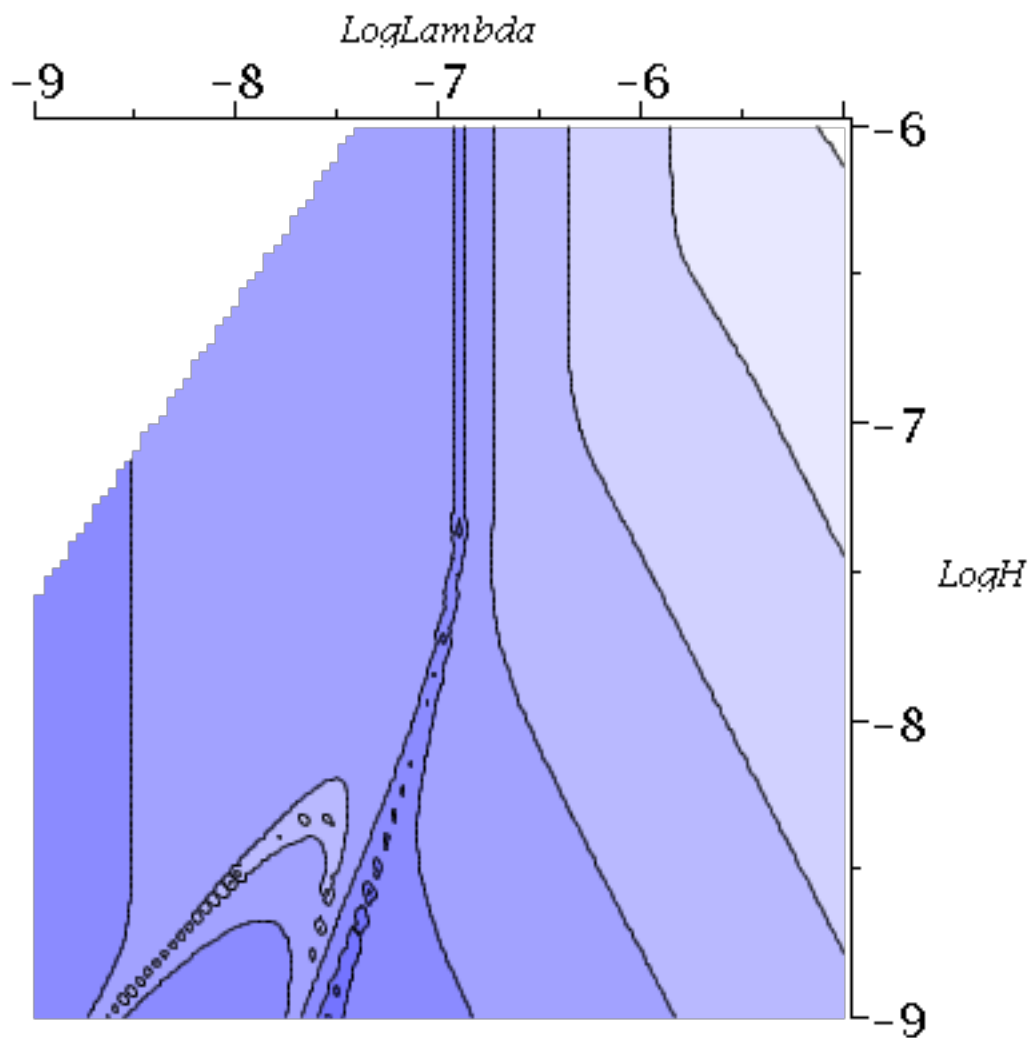


$T0 := 1900e0$

1900.

(19)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), LogLambda = -9
    ..maxLogLambda, LogH = -9 ..maxLogH, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```



$$E_{\text{photon}} := \frac{\hbar \cdot \omega}{e}$$

$$\frac{\hbar \omega}{e}$$

(20)

$$\omega := \frac{2 \cdot \pi \cdot c}{\lambda}; \hbar := \frac{h_{\text{Planck}}}{2 \cdot \pi}; e := 1.6 \text{e} - 19$$

$$\frac{2 \pi c}{\lambda}$$

$$\frac{1}{2} \frac{h_{\text{Planck}}}{\pi}$$

$$1.6 \text{e} - 19$$

(21)

$$\lambda := 1025 \text{e} - 9; h_{\text{Planck}} := 6.63 \text{e} - 34; c := 3 \text{E} 8$$

$$0.000001025$$

$$6.63 \cdot 10^{-34}$$

$$3 \cdot 10^8$$

(22)

$$E_{gap} := 9; \text{NumberOfPhotons} := \frac{E_{gap}}{E_{photon}}$$

$$9$$

$$7.420814480$$

(23)

C

$$1.850 \cdot 10^6$$

(24)

$$L_{diff} = eval\left(2 \cdot \sqrt{\frac{\kappa(T)}{C}} \cdot 1e-3, T = 300\right) \cdot 1E6$$

$$L_{diff} = 74.47264524$$

(25)

Role of temperature for a given molten depth?

$$T0 := 'T0' \# 1900e0$$

$$T0$$

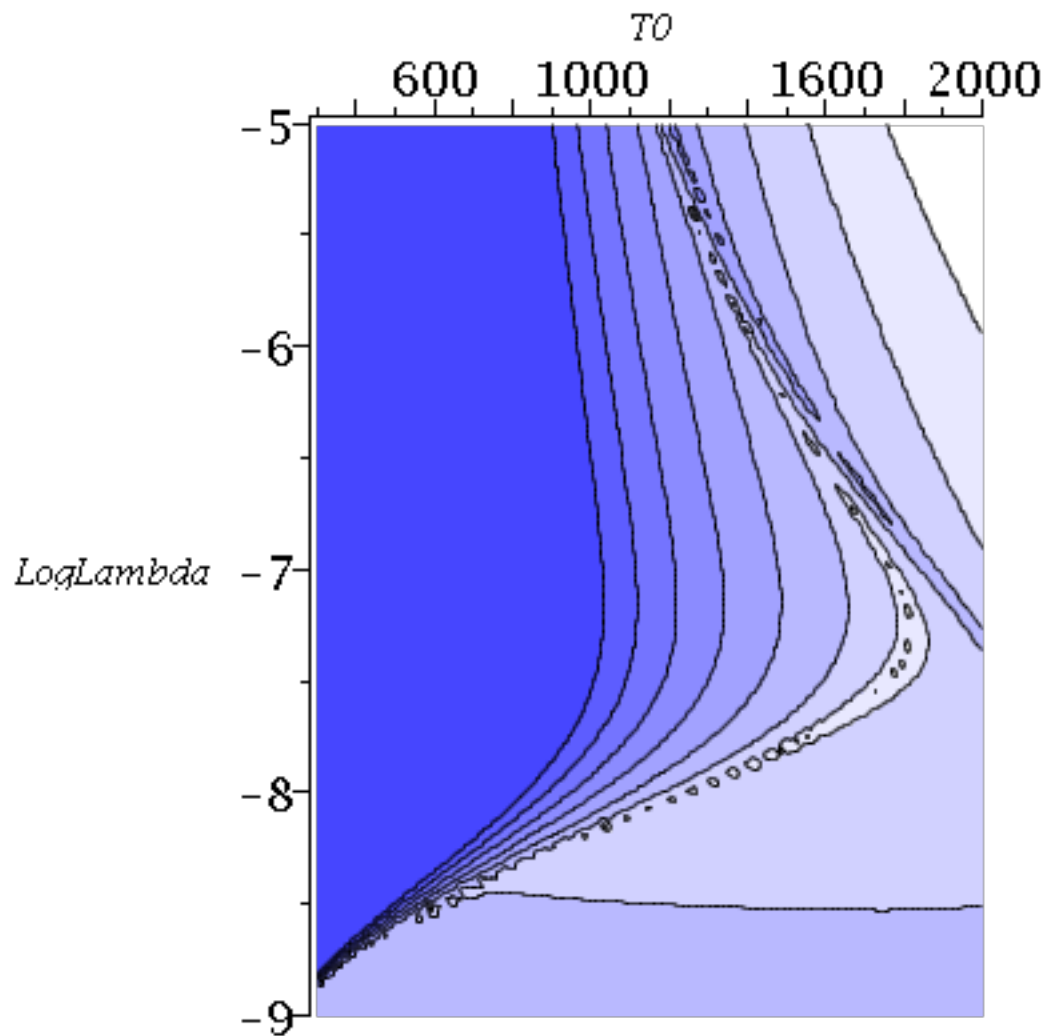
(26)

$$h := 10e-9$$

$$1.0 \cdot 10^{-8}$$

(27)

with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), T0 = 300..2000, LogLambda = -9..maxLogLambda, grid = [resolution, resolution], filledregions = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white, blue], font = [TIMES, 14, REGULAR])

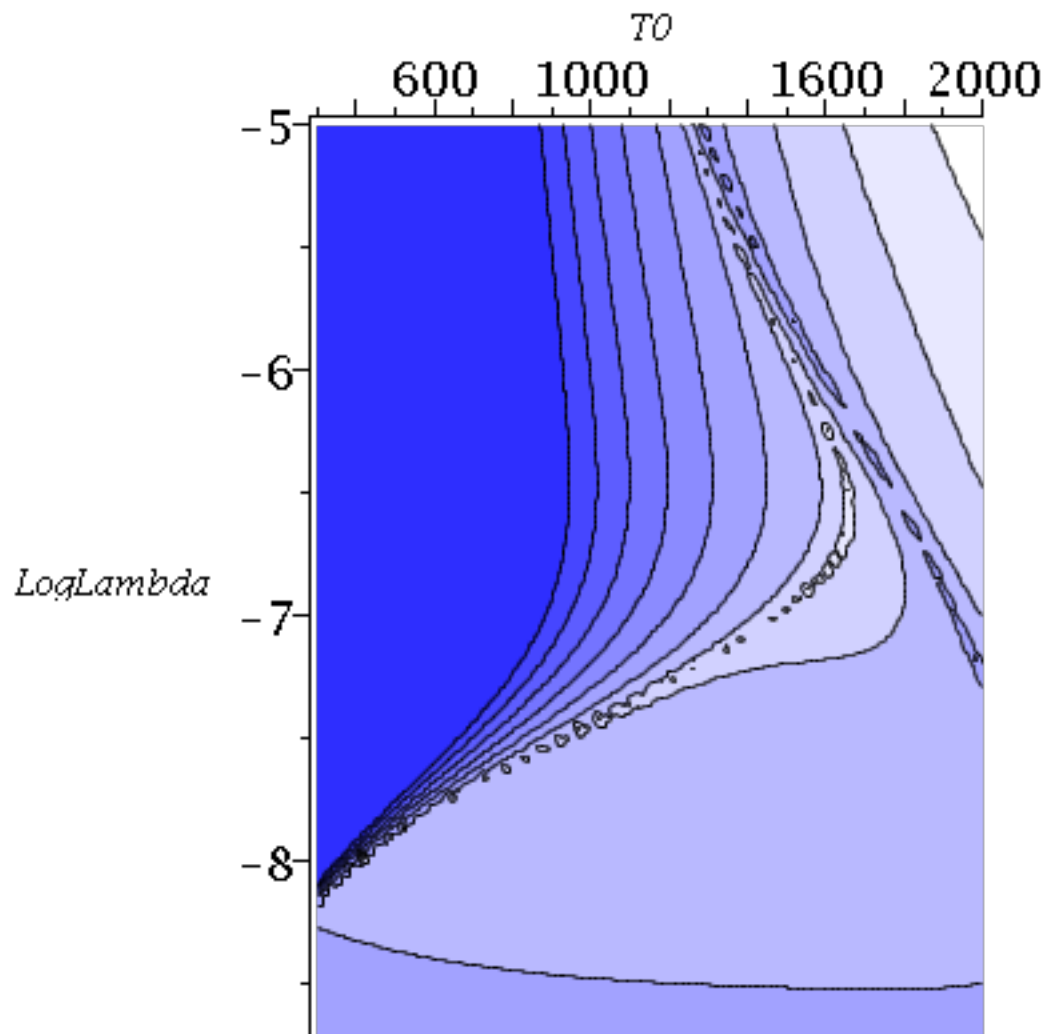


$h := 50\text{e-}9$

$5.0 \cdot 10^{-8}$

(28)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), T0 = 300..2000,
  LogLambda = -9..maxLogLambda, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```

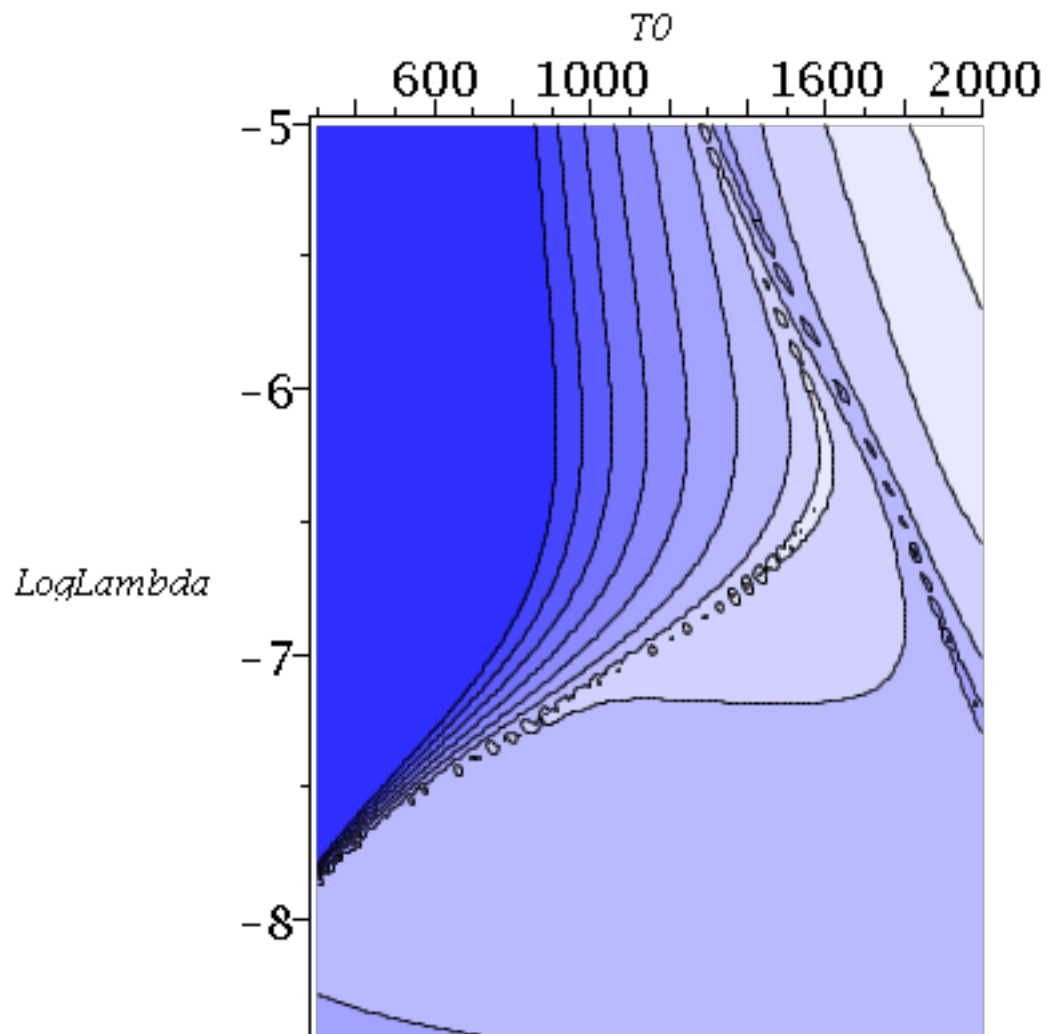


$h := 100\text{e}-9$

$1.00 \cdot 10^{-7}$

(29)

```
with(plots) : contourplot(log10(abs(eval(period2(T), T = T0))), T0 = 300..2000,
  LogLambda = -9..maxLogLambda, grid = [resolution, resolution], filledregions
    = true, contours = [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20], coloring = [white,
    blue], font = [TIMES, 14, REGULAR])
```



Non-dimensional number analysis

$Pout := 1E4; L := 1E-6$

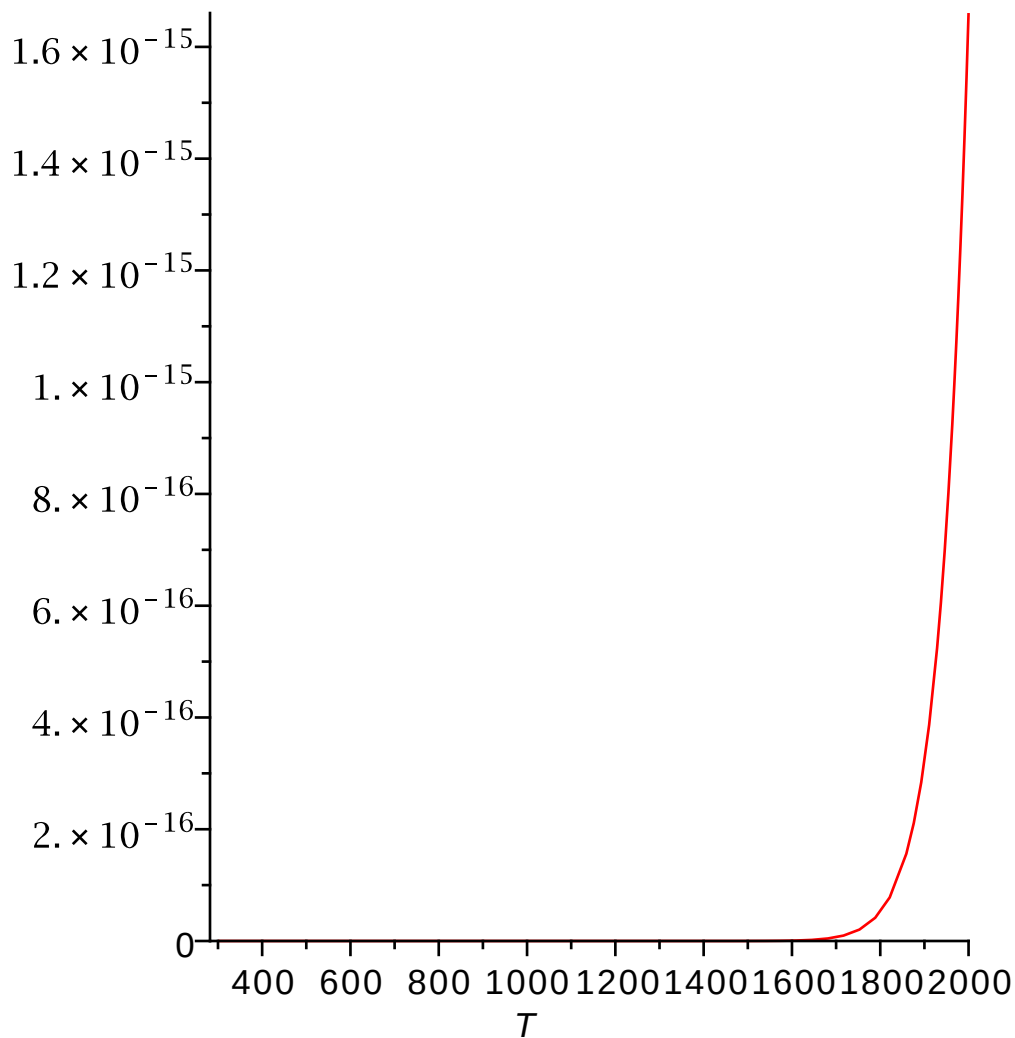
10000.

0.000001

(30)

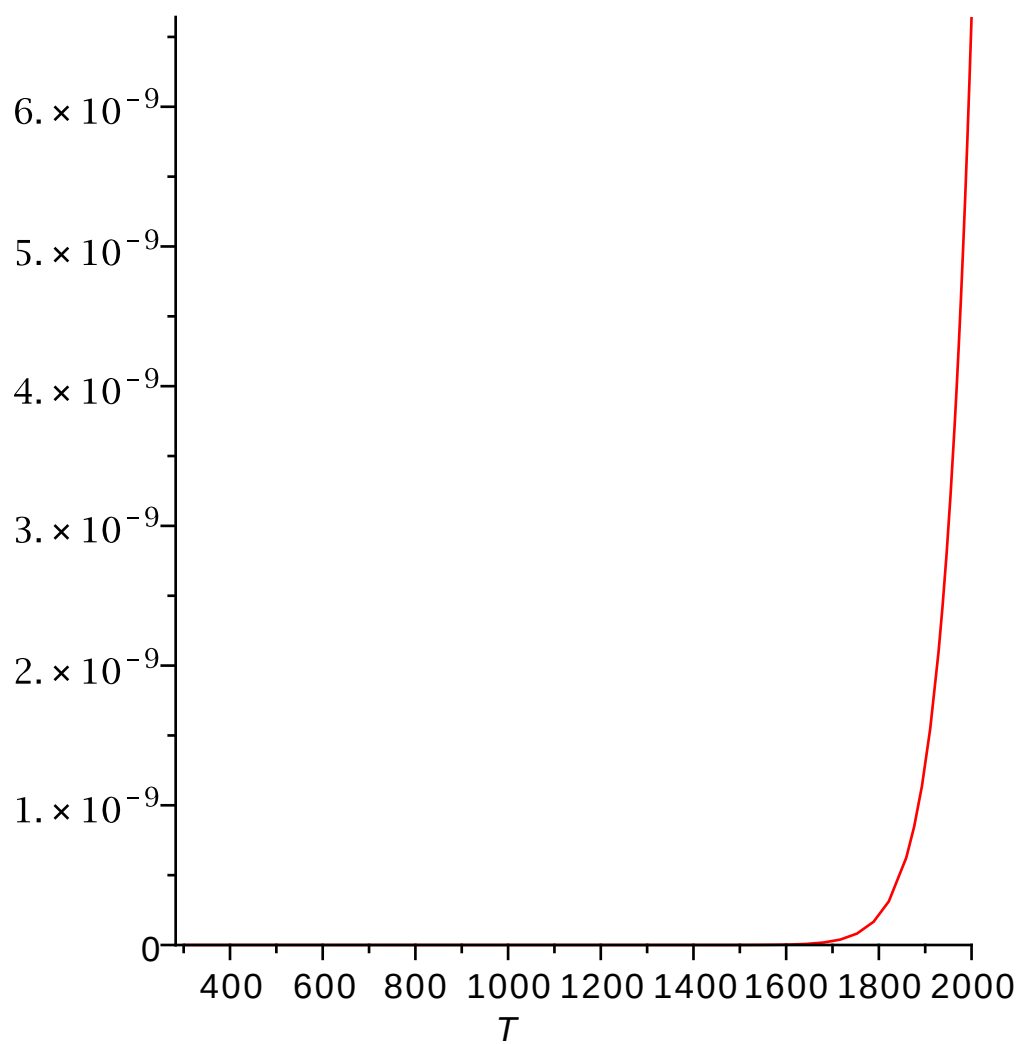
$DiffusionTime := \frac{\rho \cdot L^2}{\eta}; plot(DiffusionTime, T = 300..2000)$

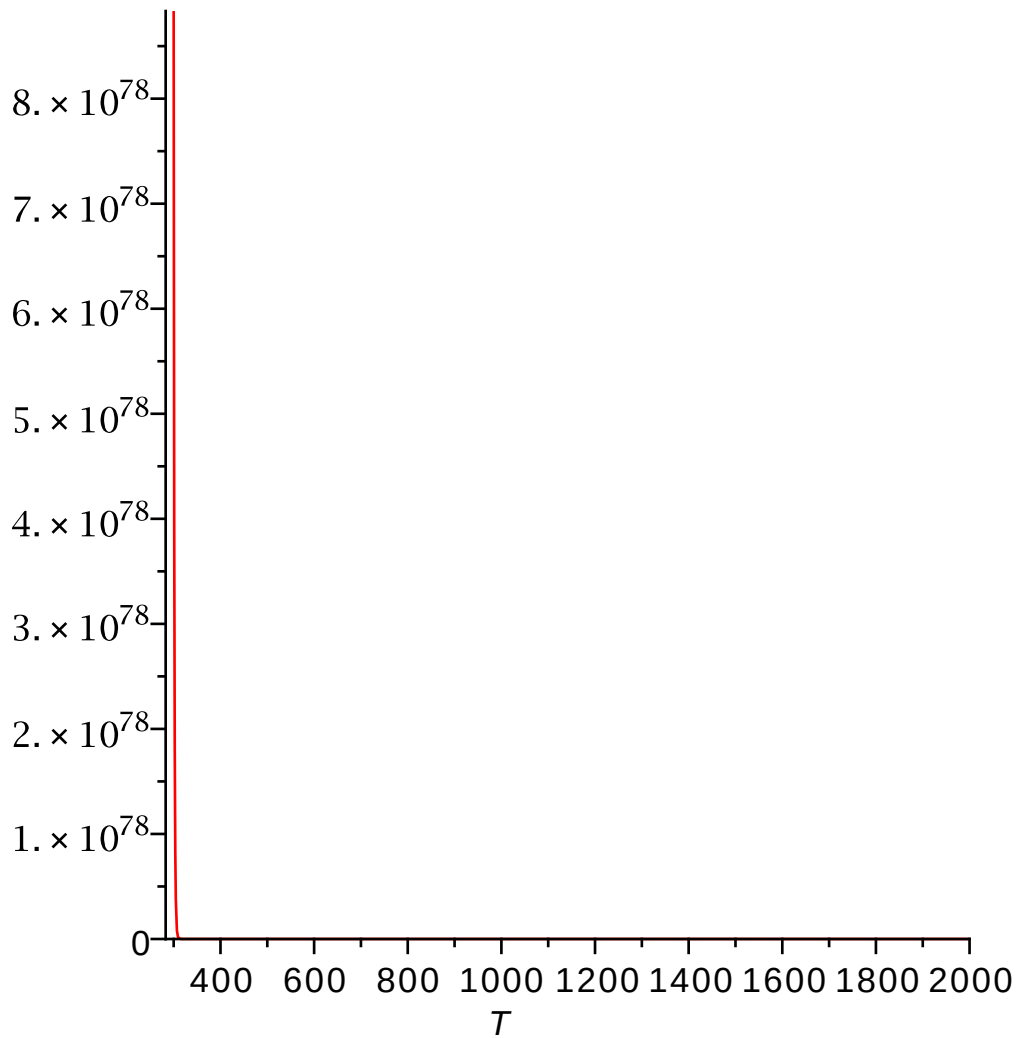
$\frac{2.500000000 \cdot 10^{-8}}{e^{\frac{62388.}{T} - 14.6668}}$



$VelocityPressure := \frac{Pout \cdot L}{eta}; TimePressure := \frac{L}{VelocityPressure}; plot(VelocityPressure, T$
 $= 300..2000); with(plots) : logplot(TimePressure, T = 300..2000)$

$$\begin{aligned}
 & \frac{0.1000000000}{e^{\frac{62388.}{T} - 14.6668}} \\
 & 0.00001000000000 e^{\frac{62388.}{T} - 14.6668}
 \end{aligned}$$





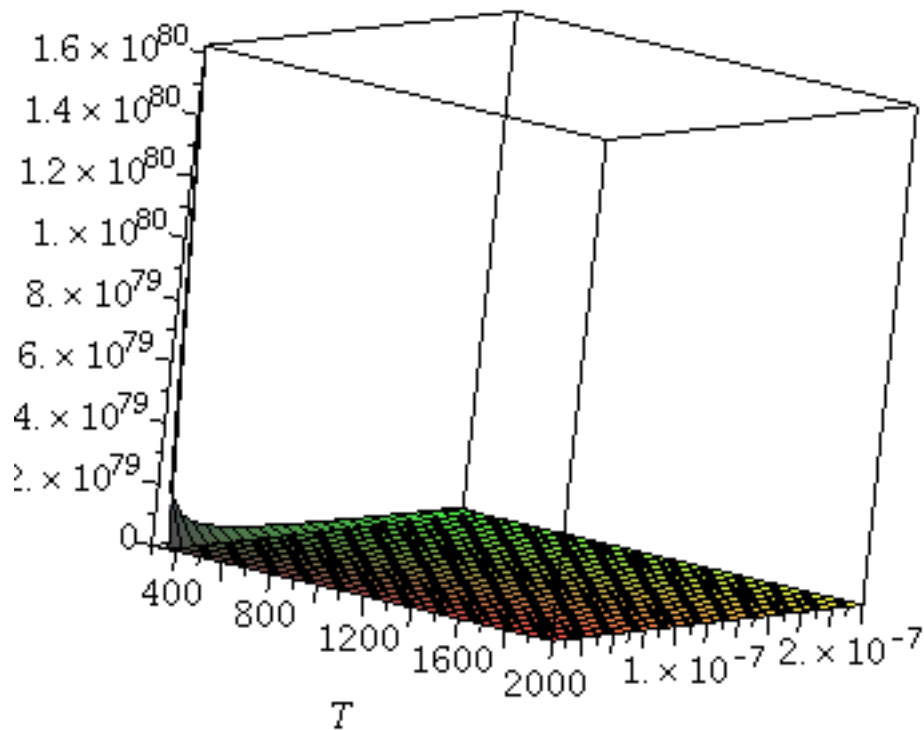
$$VelocitySurfaceTension := \frac{2 \cdot \sigma(T) \cdot h}{L \cdot \eta}; TimeSurfaceTension$$

$$:= \frac{L}{VelocitySurfaceTension}; plot3d(TimeSurfaceTension, T = 300..2000, h = 1E-9$$

$$..2E-7)$$

$$\frac{2.000000000 \cdot 10^7 (0.267 + 0.0000154 T) h}{e^{\frac{62388.}{T} - 14.6668}}$$

$$\frac{5.000000000 \cdot 10^{-14} e^{\frac{62388.}{T} - 14.6668}}{(0.267 + 0.0000154 T) h}$$



h

$$VelocityNonLinear := \sqrt{\frac{Pout}{rho}}; TimeNonLinear := \frac{L}{VelocityNonLinear}$$

$$2.000000000$$

$$5.000000000 \cdot 10^{-7}$$

(31)

$$VelocityNonLinearSurfaceTension := \sqrt{\frac{2 \cdot \sigma(T) \cdot h}{rho \cdot L^2}};$$

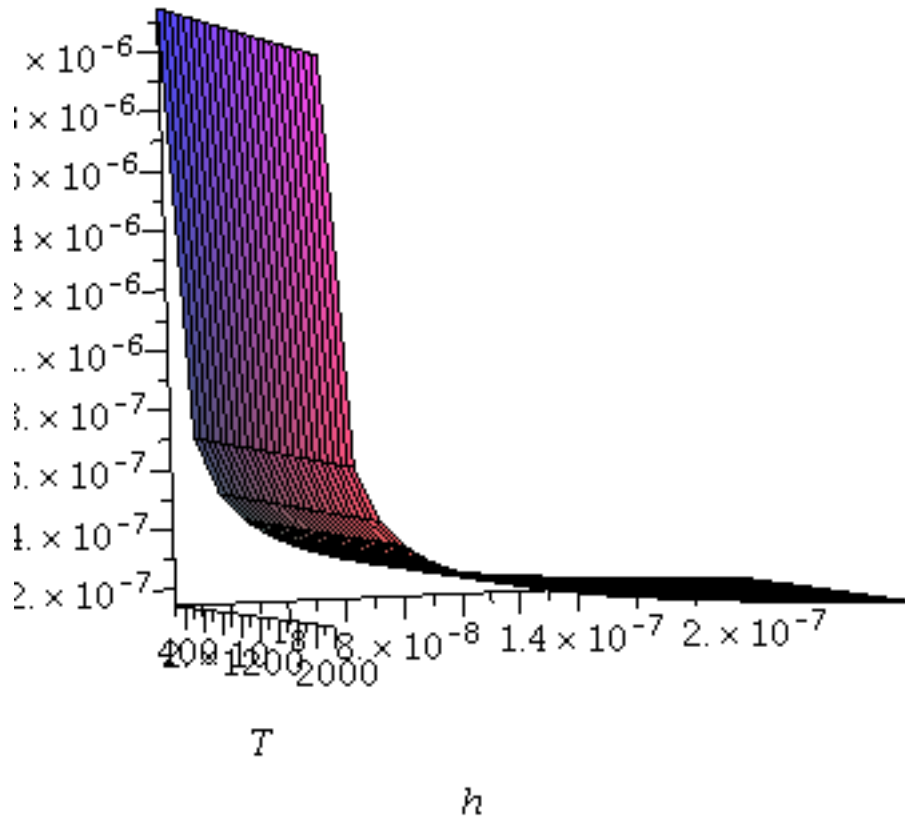
$$TimeNonLinearSurfaceTension := \frac{L}{VelocityNonLinearSurfaceTension};$$

$$plot3d(TimeNonLinearSurfaceTension, T = 300..2000, h = 1E-9..200E-9)$$

$$\sqrt{(2.136000000 \cdot 10^8 + 12320.00000 \cdot T) \cdot h}$$

$$0.000001$$

$$\sqrt{(2.136000000 \cdot 10^8 + 12320.00000 \cdot T) \cdot h}$$



$$Reynolds := \frac{\rho \cdot Velocity \cdot L}{\eta}; Capillary := \frac{\eta \cdot Velocity}{\sigma(T)}; Webber := \frac{\rho \cdot Velocity^2 \cdot L}{\sigma(T)}$$

$$\frac{0.02500000000 \sqrt{(2.136000000 \cdot 10^8 + 12320.00000 T) h}}{e^{\frac{62388.}{T} - 14.6668}}$$

$$\frac{0.1 e^{\frac{62388.}{T} - 14.6668} \sqrt{(2.136000000 \cdot 10^8 + 12320.00000 T) h}}{0.267 + 0.0000154 T}$$

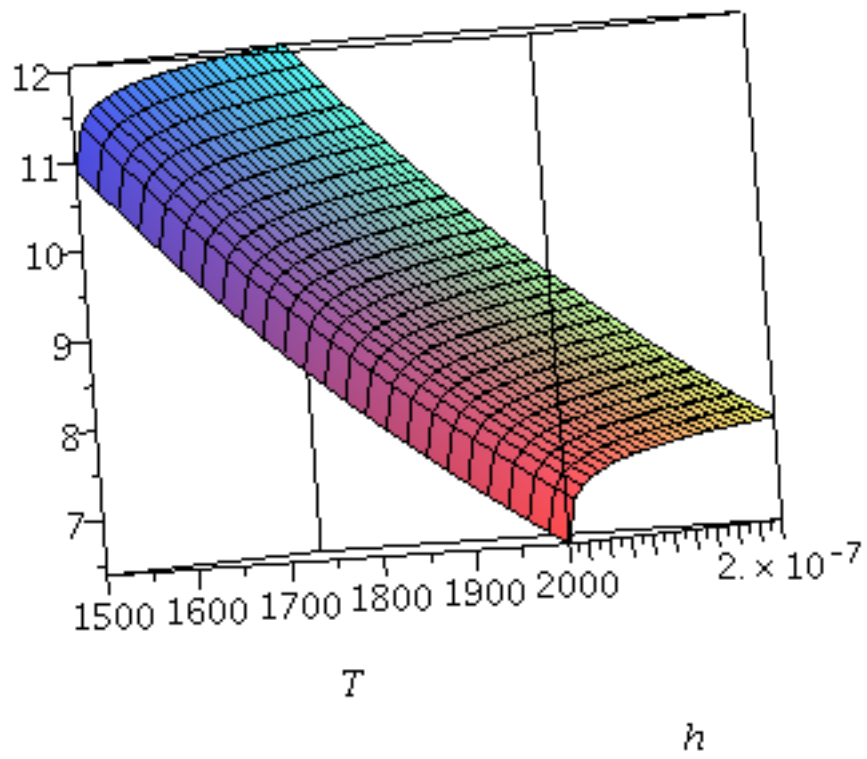
$$\frac{0.0025 (2.136000000 \cdot 10^8 + 12320.00000 T) h}{0.267 + 0.0000154 T}$$

$$Velocity := VelocityNonLinearSurfaceTension$$

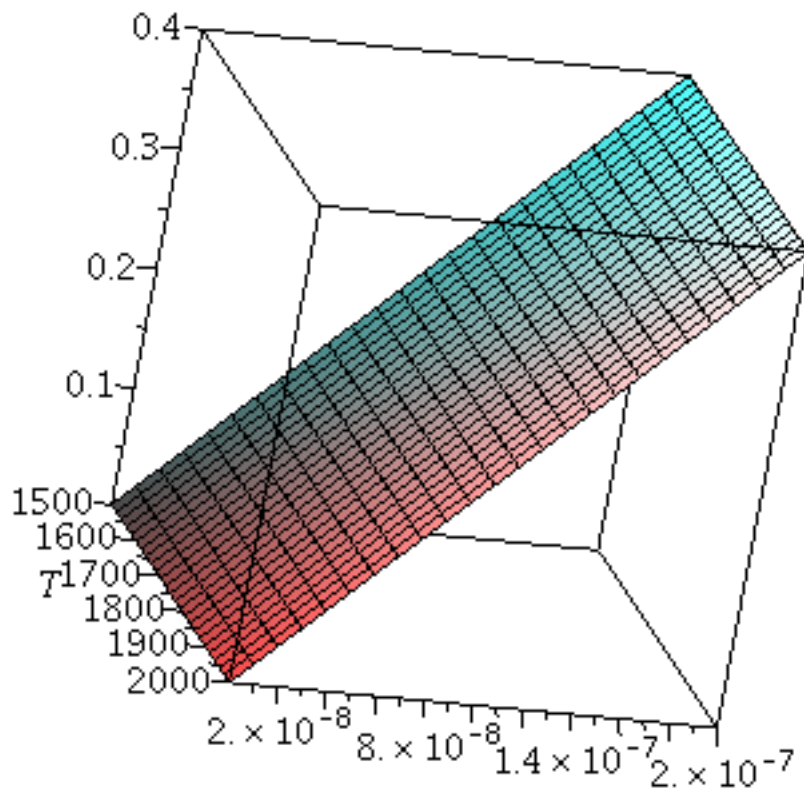
$$\sqrt{(2.136000000 \cdot 10^8 + 12320.00000 T) h}$$

(33)

$$plot3d(\log10(Capillary), T = 1500..2000, h = 1E-9..200E-9)$$



`plot3d(Webberr, T = 1500..2000, h = 1E-9..200E-9)`



```
#plot([DiffusionTime(T), TimePressure(T), TimeSurfaceTension])
```

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Estimations on the laser energy

```
lambda := 1030E-9; c := 3E8; h := 6.63e-34; e := 1.6e-19;
```

0.000001030

(34)

$$E := h \cdot \frac{c}{\text{lambda}}$$

1.931067961 10⁻¹⁹

(35)

$$\text{evalf}\left(\frac{E}{e}\right)$$

$$1.206917476 \quad (36)$$

$$E_g := 8.024234328 \cdot e$$

$$1.283877492 \cdot 10^{-18} \quad (37)$$

We make an estimation via using the Keldysh model [SPP extended theory gives]:
 $NeMax1 := 3.91105362407e+25; \#OnePulse$

$$3.91105362407 \cdot 10^{25} \quad (38)$$

$NeMax := NeMax1; N := 5; \#N \text{ pulses, } N \text{ times number of electrons}$

$$3.91105362407 \cdot 10^{25}$$

$$5 \quad (39)$$

#If considering that laser energy is summed up, instead of number of el., we get

$\#NeMax := 9e27$

$E_{max} := E_g \cdot NeMax$

$$5.021313718 \cdot 10^7 \quad (40)$$

$\rho := 2.5e3; C := 0.74E3 \cdot \rho$

$$1.850 \cdot 10^6 \quad (41)$$

$E_{max} \# J \text{ per } m^3$

$$5.021313718 \cdot 10^7 \quad (42)$$

$T_{max} := \frac{E_{max}}{C}$

$$27.14223631 \quad (43)$$

The absorbed energy is comprised between 5 times the quantity of electrons absorbed for 1 pulse, and 1 times the quantity of electrons generated by 5 times the laser energy.

$T0 := 273.15 + 20 + 24.63 + 25.17 + 25.80 + 26.46 + T_{max}$

$$422.3522363 \quad (44)$$

Just checking a dispersion relation.

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$eq1 := (\tanh(k \cdot h))$

$$\tanh(k h) \quad (45)$$

using exponentials

$$\frac{e^{k h} - e^{-k h}}{e^{k h} + e^{-k h}} \quad (46)$$

$eq12 := \frac{e^{k h} - e^{-k h}}{e^{k h} + e^{-k h}}$

$$\frac{e^{k\,h}-e^{-k\,h}}{e^{k\,h}+e^{-k\,h}} \tag{47}$$

$$simplify\Big(\frac{eq12}{\exp(k\cdot h)}\Big) \tag{48}$$

$$\frac{(e^{k\,h}-e^{-k\,h})\,e^{-k\,h}}{e^{k\,h}+e^{-k\,h}}$$

$$\omega_l:=\frac{\text{sqrt}(\exp(4\cdot k\cdot h)-1)}{(\exp(2\cdot k\cdot h)+1)}\cdot\omega_0 \tag{49}$$

$$\frac{\sqrt{e^{4\,k\,h}-1}\,\omega_0}{e^{2\,k\,h}+1}$$

$$\omega_l^2 \tag{50}$$

$$\frac{(e^{4\,k\,h}-1)\,\omega_0^2}{(e^{2\,k\,h}+1)^2}$$